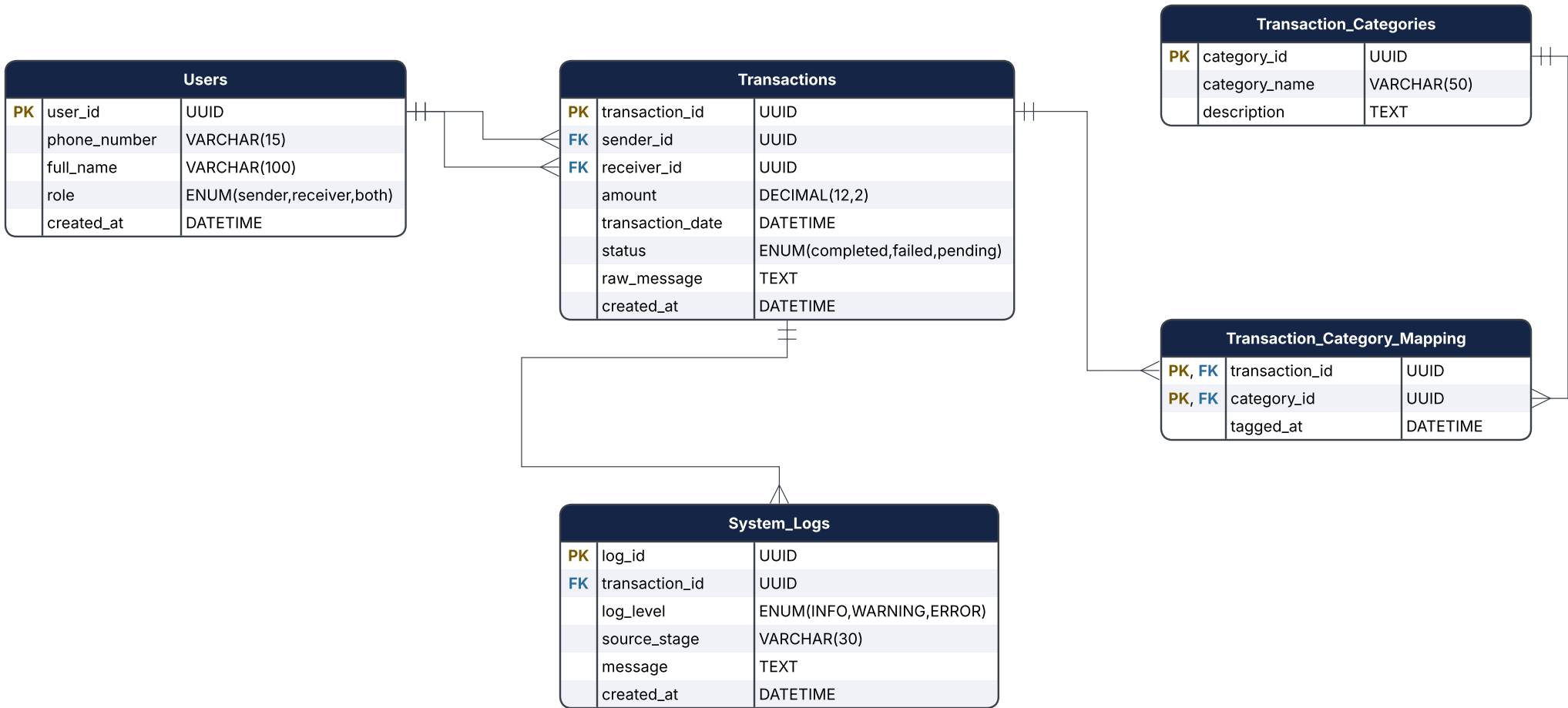




The ERD is built around five entities that map directly onto how a MoMo SMS transaction actually flows through the system. Transactions sits at the center, since every other entity exists to describe, categorize, or track it. Rather than storing sender and receiver as plain text fields, both are modeled as foreign keys pointing to a single Users entity, since a phone number can appear as a sender in one transaction and a receiver in another. This avoids duplicating user details across every transaction and keeps sender/receiver data consistent if it needs to be corrected later.

Transaction_Categories is kept separate from Transactions rather than using a fixed enum, because category types (transfers, payments, airtime, etc.) may need to be added or edited without changing the database schema. A transaction can reasonably need more than one category label at once (for example, a bill payment that is also tagged as a subscription), which is a genuine many-to-many relationship. Rather than forcing a single category_id onto each transaction, we resolved this with a junction table, Transaction_Category_Mapping, which holds paired foreign keys and its own composite key. This keeps the schema normalized instead of repeating transaction data for every category it belongs to.

Finally, System_Logs is linked to Transactions with a one-to-many relationship, since a single transaction can generate multiple log entries as it moves through parsing, cleaning, and loading. Storing logs in their own table rather than appending them to the transaction record keeps the core transaction data clean and makes it possible to trace errors back to the exact processing stage that produced them.